

Query-First Semantification of CEUR-WS Proceedings Metadata in Wikidata

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Abstract. CEUR Workshop Proceedings (CEUR-WS) has been a non-profit publisher for scientific workshop and conference proceedings since 1995. For more than 30 years, it has used HTML and PDF as the single source of truth for the metadata required by indexers such as DBLP and the German national library DNB. Scholars have long requested persistent identifiers for the core entities involved: event series, events, proceedings, papers, editors, authors, institutions, and publishers, with the goal of allowing semantic handling and querying with state-of-the-art tools.

In this work, we show how we enabled the semantification using Wikidata as the target knowledge graph infrastructure. The 2015 Semantic Publishing Challenge queries are finally permanently available with up-to-date metadata. The Scholia portal provides user-friendly and aspect-oriented access and is widely used.

We also point out the challenges of aligning the digital, social, and physical aspects of the ongoing transformation and outline a roadmap to use the available queries to bootstrap a new RDF-based metadata-first publishing approach that will mine metadata from proceedings matter and generate the HTML from it instead of the other way around.

Keywords: Knowledge Graph · Linked Data · Metadata Extraction · Publishing · Semantic Web · Wikidata

1 Introduction

Fig. 1. Most relevant entities for scientific publishing (abstract schema). Source: CR research wiki [15].

Traditional publishing workflows often handle the metadata of the required core entities (cf. Figure 1) in a manner that does not comply with the Findable,

Fig. 2. SWAT4HCLS2024 example graph walk

Accessible, Interoperable and Reusable (FAIR) principles⁴, giving low priority to these principles as well as to the Single Source of Truth (SSoT) and Single Point of Truth (SPoT) principles of data management [35]. In this work SSoT refers to the one authoritative source from which metadata originates, whereas SPoT refers to the one authoritative location at which that metadata is accessed by downstream consumers. Therefore, the opportunities to collect the metadata during the lifecycle of an event [20] are missed. As technical editors of CEUR-WS we have been feeling this pain for over a decade. This work shows how the HTML/PDF SSoT of CEUR-WS at the ceur-ws.org domain (SPoT) has been used to create a Knowledge Graph representation of the relevant entities in Wikidata between 2022 and 2026 as a step in this direction, thus allowing portals such as Scholia to present the RDF query results in an aspect-oriented and user-friendly way.

1.1 CEUR Workshop Proceedings

Since 1995 CEUR Workshop Proceedings (CEUR-WS) (<https://ceur-ws.org/>) has published more than 4,200 Volumes and over 90,000 papers as a free online service for publishing proceedings of scientific workshops (and smaller conferences) in computer science, operated by a team of unpaid volunteers.

Technically, all data and metadata of the proceedings are directly represented as a static website using HTML, PDF and HTTP and FTP protocols as provided by the workshop editors along the CEUR-WS “how to submit” guide [7].

The metadata is available after some delay of weeks or months through indexing services such as DBLP [27] and K10plus [39]⁵ and at the archive of the TIB <https://www.tib.eu/>.

After multiple unfinished attempts to make the metadata of the CEUR-WS platform available for semantic analysis and querying, we report on the successful use of Wikidata and Scholia as the knowledge graph platform to provide FAIR metadata.

The effort required was substantially higher than initially anticipated, as a number of non-trivial edge cases in the legacy metadata had to be resolved individually.

1.2 Contributions of this Paper

The core contributions presented in this work are as follows:

1. To provide the tools, infrastructure, and approaches to consistently supply high quality FAIR metadata (Semantification) for CEUR-WS as detailed in section 3.

⁴ we refer to such workflows as “unFAIR”; see Section 3.1

⁵ <https://dblp.org/>, <https://opac.k10plus.de>

2. Splitting the metadata for the three main entities: Event, Event Series, and Proceedings is a major step towards improving the metadata quality and FAIRness.
3. Providing a bootstrapping approach [3,13] to shift the single point of truth from the current HTML/PDF files to a RDF ready ontology based format as outlined in 6.

These contributions are valuable for other publishing use cases that struggle with the same pain points.

2 From Semantic Publishing Challenge 2014 via Text2KG 2023 to Production 2026

The Semantic Publishing Challenge [26,12,37] started in 2014 with Task 1 calling for “answering queries related to the quality of workshops by computing metrics from data extracted from their proceedings, also considering information about persons and events”.

The relevance of the original set of 20 queries for Task 1 was subjectively rated by us, resulting in a list sorting the queries by priority⁶. The most relevant queries and the 5 queries Q1.5, Q1.12, Q1.13, Q1.16, and Q1.17 that rely on the main index have been taken as the definition of done. We reported on the successful Wikidata export of all CEUR-WS proceeding and event metadata in 2023 at the TEXT2KG Workshop [20]. The SPARQL queries⁷ are available as named parameterized queries. The snapquery project [19] was motivated by this set of queries and was a candidate middleware for the Scholia project [43].

3 CEUR-WS Semantification in Wikidata

Figure 3 gives an overview of the CEUR-WS Semantification workflow.

3.1 Making CEUR-WS FAIR

The FAIR principles [24,42] with Persistent Identifiers (PIDs) and rich metadata as core components are being adopted by an increasing number of publishing organizations.

To make CEUR-WS metadata FAIR, a set of relevant queries should be supported, as derived from the original set of queries of the 2014 Semantic publishing challenge as outlined in Section 2. The metadata should be provided following an ontology that is based on established ontologies. The manual and automatic curation of entries should be possible with public access for all stakeholders, e.g., editors, authors, organizers, publishers, and indexers. The infrastructure should be stable and there should be sufficient trust in its long-term availability. An

⁶ https://cr.bitplan.com/index.php/List_of_Queries

⁷ https://cr.bitplan.com/index.php/Semantic_Publishing_Challenge_Queries

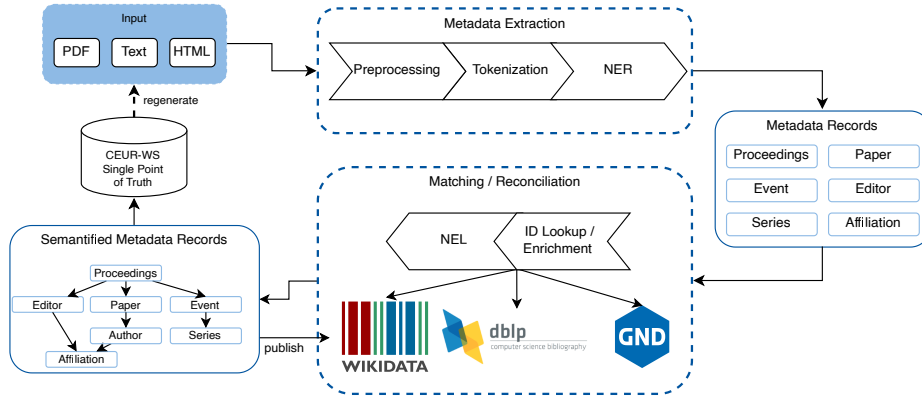


Fig. 3. Workflow of the CEUR-WS Semantification

open source non-profit infrastructure is preferred since this is also the mode of operation of CEUR-WS.

Given the HTML/PDF/text input of CEUR-WS, the corresponding Knowledge Graph needs to be created and the single-point-of-truth computer readable metadata be separated from the different representations such as HTML so that the above requirements are fulfilled.

Both the HTML and PDF encoding of the original scientific content are structured for the purpose of optimizing the display / output on paper or screens; therefore, there is a structure loss compared to what was originally available in the text document processor files that the authors might have been using [14]. Most scholars are not aware of this loss in the daily use of published content just because the documents are optimized for display and human consumption [44,2]. For the metadata extraction and use in knowledge graphs the difference is sometimes disastrous, e.g., when a simple text from a PDF document can not be extracted any more due to exotic styling and formatting or simply because only a scanned graphic image version of an older document is available that needs optical character recognition to extract the textual content.

The metadata needed for creating the knowledge graph is only available in natural language/text form and follows rules that have been changed multiple times during the history of CEUR-WS. From 2013 to 2026, there have been 33 different versions⁸ of the index file template with no proper tracing of what was changed from version to version. The index and volume HTML files were often edited manually, leading to minor undocumented differences even within these versions. The usage of the different page versions follows a long-tail Zipfian distribution with 5 versions covering 60% of all volumes.

⁸ https://cr.bitplan.com/index.php/CEUR-WS_Versions

#	Item	Volume	Acronym	Dblp	K10plus	Event	Series
4161	Proceedings of the Workshop on Artificial Intelligence and the Science of Science	Vol-4161	AIAISciSci 2025	-	k10plus	Workshop on Artificial Intelligence and the Science of Science	
4160	Proceedings of the International Scientific Workshop on Applied Information Technologies and Artificial Intelligence Systems	Vol-4160	AIT&AIS 2025	-	k10plus	International Scientific Workshop on Applied Information Technologies and Artificial Intelligence Systems	
4159	Proceedings of the 2nd International Workshop on Bioinformatics and Applied Information Technologies for medical purpose (BAITmp 2025)	Vol-4159	BAITmp 2025	-	k10plus	2nd International Workshop on Bioinformatics and Applied Information Technologies for medical purpose (BAITmp 2025)	
4158	Proceedings of the Information Technology and Implementation (ITI&I) Workshop, Artificial Intelligence Technologies and Data Science (IT&WS, AITDS 2025)	Vol-4158	ITI&WS, AITDS 2025	dblp	k10plus	Information Technology and Implementation (ITI&I) Workshop, Artificial Intelligence Technologies and Data Science (IT&WS, AITDS 2025)	
4157	Proceedings of the Workshop on AI for Evidential Reasoning	Vol-4157	AIAEVR 2025	-	k10plus	Workshop on AI for Evidential Reasoning	

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Fig. 4. CEUR Volume Browser wikidata synchronization with downstream dblp and k10plus coverage

3.2 Extracting metadata from HTML and PDF targeting Wikidata

The pyCEURmake [17] tool was created to parse the main HTML index as well as the individual proceedings HTML pages vastly improving on the earlier simple scraper by Nielsen [29]. The results have been cached in JSON and SQLite storage for quick access. Location entries have been consolidated using the geograpy3 library [36]. Author, Editor and Paper details have been aligned with the DBLP SPARQL endpoint.

Cermine [38] and GROBID [28] have been run on the papers to provide additional metadata references. Figure 4) shows a user friendly presentation of the data mined by these steps.

For each core entity, we analyzed the availability of Wikidata properties by using our wdgrid analysis tool [16] [22]. This tool also generates naive or sophisticated SPARQL queries. We aligned the queries with the stated needs of the Semantic Publishing Challenge.

snapquery [19] was used to make sure that the queries are compatible with different endpoints and do not only run on the official blazegraph based Wikidata query endpoint.

The ceur-spt prototype [18] demonstrates the benefits of semantification by improving linking and navigation capabilities compared to the current static CEUR-WS HTML pages. A complete Cermine run has been integrated so that paper-level metadata (authors, affiliations, references) is available in the per-volume pages as illustrated for Vol-3262 in Figure 5.



Fig. 5. Vol-3262 ceur-spt page: structured paper-level metadata derived from the HTML index and PDF mining pipeline.

4 Challenges in Aligning the Digital, Social and Physical Aspects

CEUR-WS is a digital-native, online-only publisher operated by volunteer editors, and serves a scholarly community of editors, authors and research institutions. Downstream indexing activities are handled by DBLP, the German National Library (DNB), TIB, Wikidata/WikiCite and the Scholia community. This section describes the challenges of aligning the digital, social, and physical aspects, since traditional proceedings are still printed on paper and stored on book shelves in libraries.

The three-way alignment is necessary to avoid cognitive dissonance in the discussions and decisions that are needed to support the metadata-first roadmap outlined in Section 8.

4.1 Digital: CEUR-WS as a native online publisher

CEUR-WS has always been online only; there is no physical artefact and there never was. As discussed in Section 3.1, the current Single Point of Truth encodes metadata in HTML and PDF optimized for printing and human display rather than for machine consumption. There has been good reason never to run any software such as a triplestore - static content is much more stable and safe than any infrastructure that needs maintenance. The Wikidata export makes it possible to query measurable metrics such as coverage and consistency. The

⁹public synchronization dashboard is an example that makes the downstream indexing observable see Figure 4.

4.2 Social: volunteer ecosystems across institutional boundaries

CEUR-WS editors, DBLP curators, DNB and TIB librarians, Wikidata and WikiCite contributors, and Scholia maintainers all operate as volunteer or mixed-funding communities, each with their own goals and procedures. Coordination across such communities is loosely coupled and based on the trust that what worked in the past will continue to do so in the future. Our work shows that there are gaps; for example, DBLP is currently indexing only about 80 % of all volumes. Trust in Wikidata as an openly editable knowledge graph has yet to be established, and active checks are needed to detect whether the metadata has been tampered with in unacceptable ways.

4.3 Physical: library catalogues and the shelf model

Library catalogues such as the DNB curated integrated authority file GND¹⁰ historically optimize for physical books — proceedings that sit on a shelf. For a digital-native publisher there is a mismatch: the distinction between *event*, *event series* and *proceedings*, which the core-entity schema of this paper treats as three separate first-class entities (Figure 1), is not covered adequately in library catalogues. Our effort to convince the DNB to systematically curate event series based on an existing concept in its ontology has so far been unsuccessful.

5 In-Use: Community Impact - WikiCite, Scholia, CEUR-WS

WikiCite is an initiative to create a bibliographic database based on Wikidata [40].

Scholia [31] is a portal serving wikidata content for the needs of scholars.

WikiCite, Wikidata, Scholia and the CEUR-WS stakeholder community are overlapping communities. There are over 100,000 distinct authors and over 6,500 distinct editors with digital traces in the DBLP CEUR-WS index entries.

5.1 Scholia

It is based on a set of over 380 named parameterized SPARQL queries organized around 40 different aspects. The Wikidata Scholarly Articles Subgraph Analysis [1] showed that roughly half of the 17 billion triples that form the main graph belong to scholarly articles and their linked entities. The Wikimedia foundation needed to perform a graph split due to a 4 Terabyte maximum storage limit of

⁹ <https://cvb.wikidata.dbis.rwth-aachen.de/wikidatasync>

¹⁰ https://www.dnb.de/EN/Professionell/Standardisierung/GND/gnd_node.html

Table 1. Scholia SWAT4HCLS 2024 panels based on SPARQL queries

#	Panel/Query	style	entities
1	data	header	event
2	people	table	scholar
3	co-authors	graph	scholar
4	sponsors	table	institution
5	timeline	timeline	event
6	topic-scores	table	
7	uses	table	
8	recent-publications	table	paper, scholar
9	proceedings	table	proceedings, paper
10	presentations	table	
11	related-events-timelocation	table	event
12	related-events-people	table	event,scholar

the Blazegraph backend in use. The Scholia community decided to use QLever [4] as the new backend and make all queries SPARQL1.1 compliant [43]. Note that before the graph split there was a hold on adding more scholarly articles to Wikidata which was the reason we did not mass-insert CEUR-WS paper meta-data yet. Writes to Wikidata are performed through the officially approved bot account *User:CEUR-WS* [8], for which the community permission was obtained in 2023 [41].

Table 1 shows 12 SPARQL-driven panels on the Scholia event aspect [31]. All queries are enabled by this work and rendered identically by the legacy Blazegraph endpoint at <https://scholia.toolforge.org/event/Q124669329> and by the post graph-split QLever endpoint [4,43] at <https://qllever.scholia.wiki/event/Q124669329>, illustrated for SWAT4HCLS 2024 (Q124669329)¹¹. All 12 panels survived the 2025 Wikidata graph split unchanged; only the underlying engine and the SPARQL dialect (1.1 compliant) differ.

Figure 6 shows the user-friendly display of the coauthorship as graph visualization.

5.2 WikiCite

WikiCite [40] provides the bibliographic backbone on which CEUR-WS meta-data is contributed to Wikidata. Since the CEUR-WS Wikidata bot was activated the community members contributed to the bot created entries.

Figure 7 shows the distribution of Wikidata editors who have created or revised CEUR-WS *proceedings* items (conference proceedings (Q1143604)¹²). The long-tail distribution illustrates that while the officially approved *User:CEUR-WS* bot [8] accounts for the bulk of systematic mass contributions, a broad

¹¹ <https://www.wikidata.org/wiki/Q124669329>

¹² <https://www.wikidata.org/wiki/Q1143604>

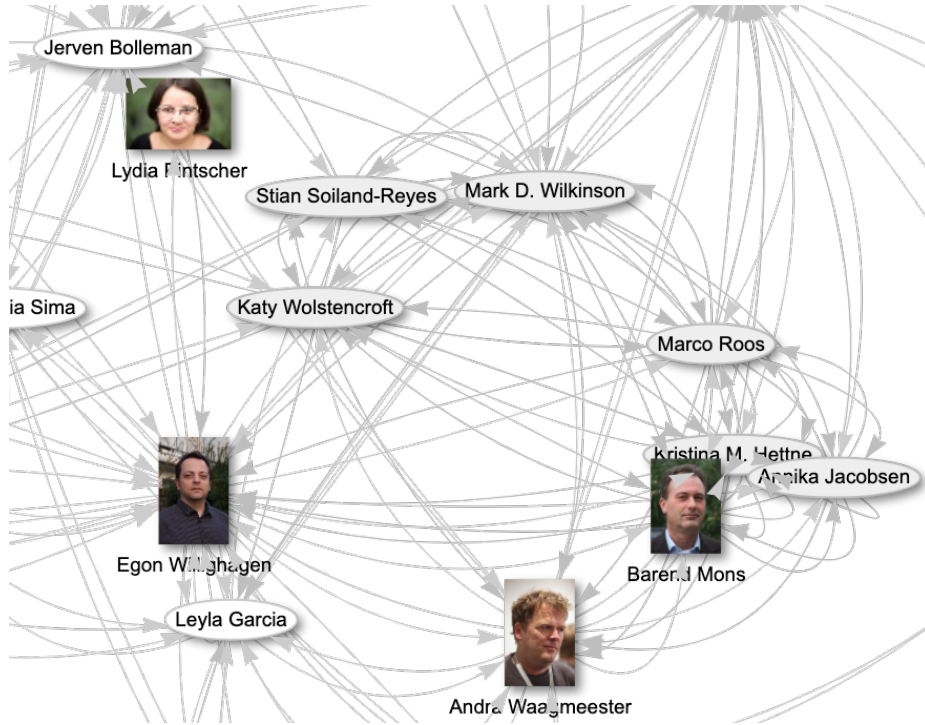


Fig. 6. SWATH4HCLS 2024 Coauthor graph visualization

community of individual WikiCite editors continuously curates, corrects and enriches the proceedings metadata. This overlap of bot-driven bulk ingestion and human community curation is a core strength of the WikiCite approach and a prerequisite for the Scholia panels discussed in Section 5.1 above. The plot was generated with `ceur-wd-contributions` [17] by traversing the revision history of all CEUR-WS proceedings items on Wikidata.

5.3 CEUR-WS stakeholders

The CEUR-WS bot [8] currently exports *proceedings* and *event* items only. Items for the related entities event series, paper, scholar and institution are created and edited by the CEUR-WS stakeholders that are active in the Wikidata community. The SWAT4HCLS 2024 case has already been presented in Section 1 (graph walk, Figure 2) and in this Section 5 (Scholia panels, Table 1 and Figure 6).

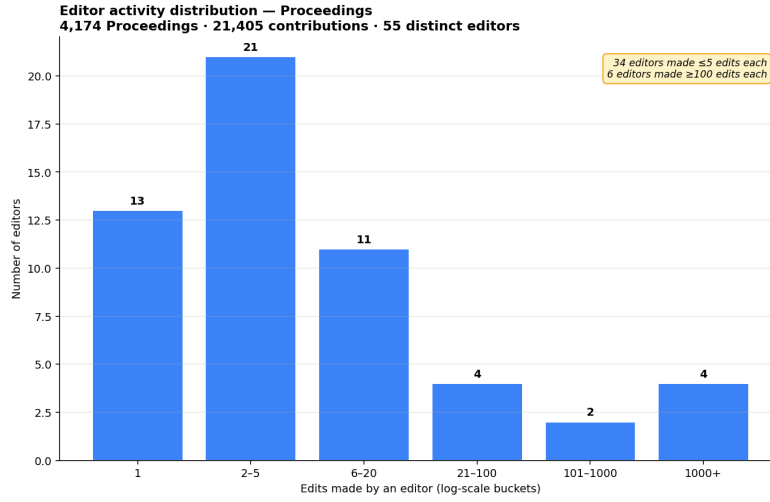


Fig. 7. Distribution of WikiCite community editors contributing to CEUR-WS proceedings items on Wikidata.

6 Single Point of Truth Bootstrapping

6.1 CEUR-WS Proceedings Ontology

The CEUR-WS Proceedings Ontology formalizes the structure of the semantified data. A first draft is available and covers all entity classes defined by the Semantification Challenge (see Figure 8). To maximize reusability and interoperability, terms from established ontologies are reused wherever appropriate, including the SPAR Ontologies [32] such as Dublin Core and Schema.org are reused, and crosswalks to Wikidata and DBLP are provided. The draft is already operational: JSON-LD representations are generated for every volume.¹³ Ongoing schema consolidation is driven through two open discussions in the public `semantify` repository, covering the schema itself [10] and sample volume subgraphs [9].

Four design questions remain open:

1. **Vocabulary selection.** Widely adopted vocabularies such as Schema.org and Dublin Core are preferred over bespoke terms, with mappings provided where necessary; the precise choice is still under discussion for several entity types.
2. **Model granularity.** Open issues include the representation of session structure and the distinction between joint proceedings and joint events.
3. **Tooling.** Generation of the JSON-LD from the workshop website and PDFs should be automated, so that editors curate only the JSON-LD and the

¹³ e.g. <http://ceurspt.wikidata.dbis.rwth-aachen.de/Vol-3264.jsonld>

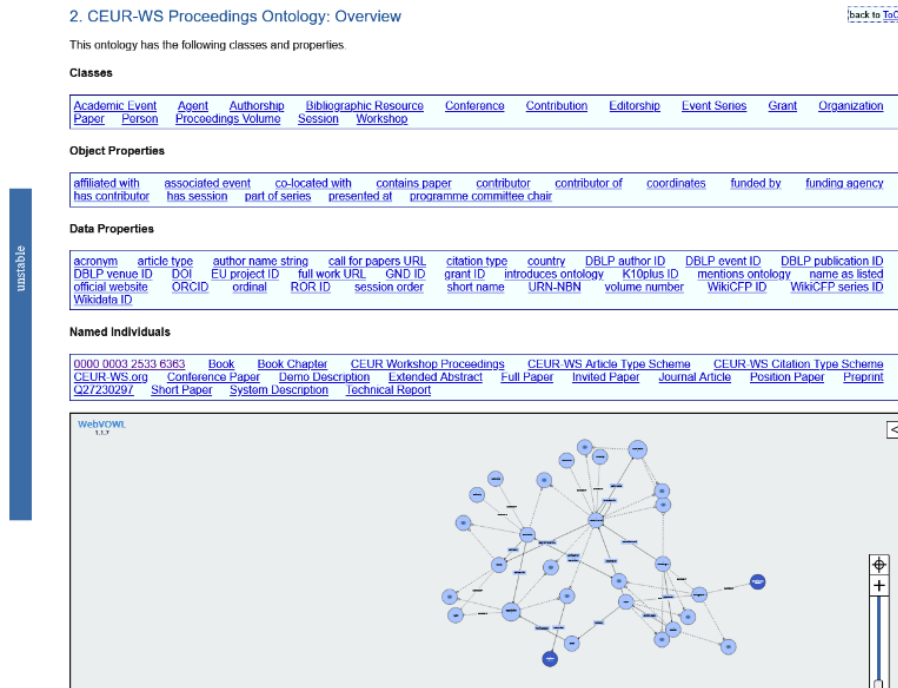


Fig. 8. Overview of the proposed CEUR-WS Proceedings ontology

HTML page is regenerated from it. This inverts the current HTML-as-SPoT situation described in Section 3.1.

4. **Shape-based validation.** SHACL [25] or ShEx [34] shapes should be generated from the ontology to encode editor experience as machine-checkable rules, including cardinalities, permitted value sets, and required identifiers. Beyond ingestion-time validation, shapes can drive form generation for the editor tooling in (iii), serve as machine-readable documentation for downstream consumers, and enable continuous quality monitoring of the published knowledge graph.

7 Lessons Learned

7.1 Queries as first-class citizens

The Semantic Publishing Challenge and the Scholia portal both use queries as the core structuring elements. Using the guiding principle “If you can not query it it does not exist” changes the view on how to provide metadata. While traditionally schemas and ontologies are created first to allow querying for use-case specific areas of interest, the query-first and query as first-class citizens

approach describes the information needs of stakeholders as queries. Having the queries as named parameterized technology independent entities themselves is helpful [19]. Nielsen’s Synia [30] approach shows a lightweight application of this idea.

A main success factor for this work was being able to apply metadata to the queries. This way the queries could be ordered by task and/or relevance and linked to their respective implementation. These links were just motivations - as first class citizens the queries are available with proper URIs as persistent identifiers and part of a meta-knowledge graph that is useful for the further semantification steps.

7.2 Property creation as a community process

Wikidata properties such as *colocated with* (P11633)¹⁴ undergo a Wikidata’s property proposal process¹⁵ process before a new property is created. As described in [20] the creation of property proposal¹⁶ took some effort and led to the clarification that the property is asymmetric. The property was considered as highly relevant and well defined.

7.3 Legal aspects of Publishing Personal Data of Scholars

The proposed workflow calls for publishing personal data of authors and editors early in the life cycle of an event. According to the legal advisors we asked, this is legal since a combination of conditions (Art. 6 GDPR) is met: Explicit consent is given by the stakeholders (Art. 6(1)(a)), the legitimate interest of the stakeholders is pursued (Art. 6(1)(f)), the data might be public already, the processing is necessary for scientific research (Art. 6(1)(e), Art. 89). The strongest condition is that there might be a legal obligation (Art. 6(1)(c)) as would be the case if publishing via major print publishers that have to provide a copy of the proceedings to the German National Library according to German law. Consent, Public Data, and Contractual Necessity are crucial for scholarly publishing. The legal basis and data-subject rights for publication of personal data of scholars by CEUR-WS are documented in the CEUR-WS privacy policy¹⁷ and tracked in the project issue tracker issue #11 of the ceur semantify project¹⁸.

7.4 Incremental migration over one-shot replacement

The challenge in introducing the new CEUR-WS workflow is to find a sweet spot between effort and benefit [33]. Given the long-tail distribution of issues of ever more corner and exotic cases that have shown up in the analysis of the existing

¹⁴ <https://www.wikidata.org/wiki/Property:P11633>

¹⁵ https://www.wikidata.org/wiki/Wikidata:Property_proposal

¹⁶ https://www.wikidata.org/wiki/Wikidata:Property_proposal/colocated_with

¹⁷ <https://ceur-ws.org/privacy-dsgvo.html>

¹⁸ <https://github.com/ceurws/semantify/issues/11>

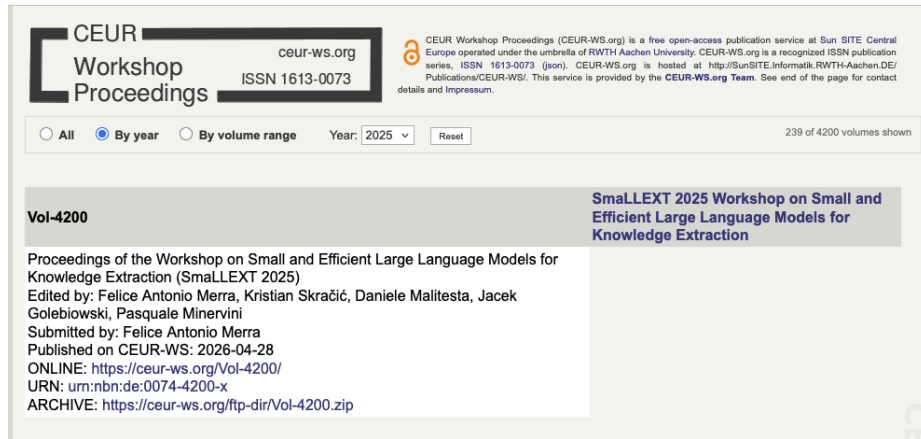


Fig. 9. Year-split prototype of the CEUR-WS index (`index_alt.html`): a step towards iterative replacement of the monolithic index page.

data and workflow there is potentially a huge list of “small” problems that would require an inadequately high amount of effort to be solved systematically. The “sweet spot” is in solving only the most relevant problems and not fixing some of the exotic details at all or doing it manually. Therefore we will avoid a “big bang” introduction, which would replace the old approach in one single big (and therefore very risky) step. Instead we propose to do smaller changes gradually. As a first step, the main `index.html` shall be split by years (with a link to the old legacy complete list). Then, the per-year list may be generated from the single point of truth data. This will be continued iteratively, going from the current year back to a point where it seems no longer reasonable to spend further effort.

A first bootstrapping prototype of this year-split is already available (see Figure 9).

8 Roadmap: Metadata first Publishing

The coverage of the core entities Event Series, Papers, Scholars and Institutions needs to be improved based on consolidating and completing the RDF data of the downstream indexing services at DBLP and German National Library (DNB). The central long-term objective of this effort is to shift the CEUR-WS single point of truth from the current HTML/PDF artefacts to a workflow in which the metadata is mined from the proceedings matter and rendered artefacts are generated from it. The 2026 migration of the CEUR-WS infrastructure from RWTH Aachen to Gesellschaft für Informatik provides the opportunity to take further concrete steps in this direction. The open-source CEUR-WS semantify project [23] coordinates this work.

8.1 Community proposals for the future workflow

Several complementary proposals have been provided during the timeframe of this work:

- *Author-driven metadata entry*: Michael Cochez proposes re-using the `ceur-ws-prototype` [11], inspired by `ceur-make-ui` [6], with a simple HTML form so that authors enter their own metadata, thus automating volume creation and reducing editor workload.
- *Automated extraction and compliance checking*: Jerven Bolleman proposes automating metadata/PDF extraction, index generation and CEUR compliance checks using `ceur-index-html-helper` [5], already applied to generate <https://www.jerven.eu/swat4hcls-proceedings-2026/>.
- *SPT-first research prototype*: the authors propose making the `ceur-spt` backend fully functional first and only then adding the frontend, following the `SemPubFlow` prototype approach.
- *Historical context*: the 2019 XML-based proposal by Sarven Capadisli and the ESWC 2022 background discussion by Nathanael Arndt are acknowledged as previous ideas and work.

9 Conclusion

CEUR-WS proceedings metadata is systematically made available in Wikidata via the CEUR-WS bot. The relevant Task 1 Named Parameterized Queries of the Semantic Publishing Challenge of 2014/2015 are now available on different endpoints namely the SPARQL1.1 compatible QLever endpoint. The Scholia portal provides production access to queries based on and extending this set. The Wikidata, WikiCite, Scholia and CEUR-WS stakeholder communities have appreciated and contributed to this work.

9.1 Limitations

The contributions reported here are subject to several limitations. First, Wikidata coverage is currently complete for the *proceedings* and *event* entities only; items for papers, scholars, institutions and event series rely on complementary community curation and are therefore not uniformly complete. Second, the reconciliation pipeline depends on the availability and quality of external endpoints (DBLP, K10plus, geograpy3); degradation or schema changes in these sources propagate into the pipeline. Third, roughly 20 % of CEUR-WS volumes are not yet indexed by DBLP, which limits editor and author disambiguation for that subset. Fourth, the current Single Point of Truth remains the legacy HTML/PDF representation; the metadata-first workflow outlined in Section 8 has not yet replaced it in production. Finally, generalisation beyond CEUR-WS to other publishers has not been empirically evaluated and is left for future work.

9.2 Supplemental Material Statement

Source code for the prototypes and semantification tools is available via the repositories of the github ceurws organization¹⁹. Extended supplemental material — including the schema diagrams (Figures 1 and 2) in their editable Graphviz form, the SPARQL queries derived from the Semantic Publishing Challenge, and background research — is provided by the CR research wiki [15]. The LaTeX sources, bibliography, build pipeline and the `AGENTS.md` governance policy for AI-assisted editing of this paper are available at [21]. The cited software artefacts are released under Apache-2.0: `ceur-spt` [18], `pyCEURmake` [17], `snapquery` [19], and `wdgrid` [16].

9.3 Acknowledgements

We would like to thank Jakob Voß for helping with the k10plus matching and creating the Wikidata property colocated with (P11633)²⁰ in due time.

Thomas Hoeren and Jonas Kuiter of ITM, Münster kindly gave hints for legal aspects as summarized in section 7.3.

This paper is dedicated to the memory of CEUR-WS board member Ralf Klamma † January 2023.

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9.4 Declaration of use of Generative AI

The authors used AI-based coding assistants during the engineering and preparation of this work. Use of such assistants is governed by an `AGENTS.md` policy file maintained in this paper’s repository and, equivalently, in the source repositories of the software artefacts `ceur-spt` [18] and `pyCEURmake` [17]. The policy mandates:

- a plan-and-confirm protocol: the assistant must state its analysis, plan, and scope, and receive explicit human confirmation before any read, write, or execution;
- a strict scope for the manuscript itself: AI assistance is limited to spelling/grammar corrections and formatting support (tables, figures, build automation). AI was not used to generate, rewrite or contribute original scientific content, arguments, or prose;

All scientific claims, interpretations and conclusions are the sole responsibility of the authors.

¹⁹ <https://github.com/ceurws>

²⁰ <https://www.wikidata.org/wiki/Property:P11633>

²¹ ConFIdeNT project; see <https://gepris.dfg.de/gepris/projekt/426477583>

References

1. AKhatun: Wikidata Scholarly Articles Subgraph Analysis, https://wikitech.wikimedia.org/wiki/User:AKhatun/Wikidata_Scholarly_Articles_Subgraph_Analysis
2. Ansonge, L.: Let's publish full-text scientific articles in html, not just pdf. *European Science Editing* **47** (Nov 2021). <https://doi.org/10.3897/ese.2021.e75834>
3. Bardini, T.: Bootstrapping: Douglas Engelbart, Coevolution, and the Origins of Personal Computing. Stanford University Press, USA (2001)
4. Bast, H., Buchhold, B.: QLever. In: *Proceedings of the 2017 ACM Conference on Information and Knowledge Management*. pp. 647–656. ACM (Nov 2017). <https://doi.org/10.1145/3132847.3132921>
5. Bolleman, J.: `ceur-index-html-helper`. <https://github.com/JervenBolleman/ceur-index-html-helper>
6. CEUR-WS community: `ceur-make-ui`. <https://github.com/ceurws/ceur-make-ui>
7. CEUR-WS Editor Team: How to submit to ceur workshop proceedings. <https://ceur-ws.org/HOWTOSUBMIT.html> (2019)
8. CEUR-WS Editor Team: Wikidata bot account: User:CEUR-WS (2023), <https://www.wikidata.org/wiki/User:CEUR-WS>
9. CEUR-WS semantify community: Sample volume subgraphs (discussion #15). <https://github.com/ceurws/semantify/discussions/15> (2026)
10. CEUR-WS semantify community: Schema consolidation (discussion #6). <https://github.com/ceurws/semantify/discussions/6> (2026)
11. Cochez, M., contributors: `ceur-ws-prototype`. <https://github.com/ceurws/ceur-ws-prototype>
12. Di Iorio, A., Lange, C., Dimou, A., Vahdati, S.: Semantic publishing challenge – assessing the quality of scientific output by information extraction and interlinking. In: Gandon, F., Cabrio, E., Stankovic, M., Zimmermann, A. (eds.) *Semantic Web Evaluation Challenges*. pp. 65–80. Springer International Publishing, Cham (2015)
13. Engelbart, C.: About Bootstrapping. <https://www.dougenelbart.org/content/view/226/269/> (2007)
14. Fahl, W.: The history of scientific publishing (2023), https://cr.bitplan.com/index.php/The_History_of_Scientific_Publishing
15. Fahl, W.: CR: Semantic MediaWiki on semantification, knowledge graphs and scientific publishing. https://cr.bitplan.com/index.php/Main_Page (2026)
16. Fahl, W.: `wdgrid`: NiceGUI-based Wikidata grid display and sync (2026). <https://doi.org/10.5281/zenodo.20035246>
17. Fahl, W., Holzheim, T., Ayan9801: `pyCEURmake`: Python implementation for CEUR Workshop Proceedings (2026). <https://doi.org/10.5281/zenodo.20034998>, also known as `cvb` (CEUR Volume Browser)
18. Fahl, W., Holzheim, T., byzakyz: `ceur-spt`: RESTful Single Point of Truth server for CEUR-WS (2026). <https://doi.org/10.5281/zenodo.20037981>
19. Fahl, W., Holzheim, T., Hoyt, C., Priskorn, D., et al.: SnapQuery: FAIR Management of Query Sets to Mitigate Query Rot in Knowledge Graphs (2026). <https://doi.org/10.5281/zenodo.17802424>
20. Fahl, W., Holzheim, T., Lange, C., Decker, S.: Semantification of CEUR-WS with wikidata as a target knowledge graph. In: *Joint Proceedings of Text2KG and BiKE co-located with ESWC 2023. CEUR Workshop Proceedings*, vol. 3447, pp. 205–225. CEUR-WS.org (2023), https://ceur-ws.org/Vol-3447/Text2KG_Paper_13.pdf

21. Fahl, W., Holzheim, T., Lange, C., Decker, S.: CEURWS_Semantification_ISWC2026: Paper source repository with AGENTS.md governance policy. https://github.com/WolfgangFahl/CEURWS_Semantification_ISWC2026-paper (2026)
22. Fahl, W., Holzheim, T., Westerinen, A., Lange, C., et al.: Property cardinality analysis to extract truly tabular query results from wikidata. In: Proceedings of the 3rd Wikidata Workshop 2022. CEUR Workshop Proceedings, vol. 3262. CEUR-WS.org (2022), <https://ceur-ws.org/Vol-3262/paper7.pdf>
23. Fahl, W., et al.: semantify: Semantification of the ceur-ws publication workflow. <https://github.com/ceurws/semantify> (2026)
24. GO FAIR International Support and Coordination Office: Fair principles (2019), <https://www.go-fair.org/fair-principles/>
25. Knublauch, H., Kontokostas, D.: Shapes constraint language (SHACL). Tech. rep., W3C (Jul 2017), <https://www.w3.org/TR/shacl/>
26. Lange, C., Di Iorio, A.: Semantic publishing challenge – assessing the quality of scientific output. In: Communications in Computer and Information Science, pp. 61–76. Springer International Publishing (2014). https://doi.org/10.1007/978-3-319-12024-9_8
27. Ley, M.: DBLP. Proceedings of the VLDB Endowment **2**(2), 1493–1500 (Aug 2009). <https://doi.org/10.14778/1687553.1687577>
28. Lopez, P.: GROBID: Combining automatic bibliographic data recognition and term extraction for scholarship publications. In: Research and Advanced Technology for Digital Libraries, pp. 473–474. Springer Berlin Heidelberg (2009). https://doi.org/10.1007/978-3-642-04346-8_62
29. Nielsen, F.Å.: Scraping ceur workshop proceedings, <https://github.com/WDscholia/scholia/blob/master/scholia/scrape/ceurws.py>
30. Nielsen, F.Å.: Synia: Displaying data from wikibases (2023). <https://doi.org/10.48550/ARXIV.2303.15133>
31. Nielsen, F.Å., Mietchen, D., Willighagen, E.: Scholia, scientometrics and wikidata. In: Blomqvist, E., Hose, K., Paulheim, H., Lawrynowicz, A., et al. (eds.) The Semantic Web: ESWC 2017 Satellite Events. pp. 237–259. Springer International Publishing, Cham (2017). https://doi.org/10.1007/978-3-319-70407-4_36
32. Peroni, S., Shotton, D.: The spar ontologies. In: The Semantic Web – ISWC 2018. pp. 119–136. Springer International Publishing, Cham (2018)
33. Pohl, K., Rupp, C.: Requirements engineering fundamentals. Rocky Nook, Santa Barbara, CA, 2 edn. (May 2015)
34. Prud’hommeaux, E., Labra Gayo, J.E., Solbrig, H.: Shape expressions: an rdf validation and transformation language. In: Proceedings of the 10th International Conference on Semantic Systems. p. 32–40. SEM ’14, Association for Computing Machinery, New York, NY, USA (2014). <https://doi.org/10.1145/2660517.2660523>
35. Queiroz, M., Tallon, P., Coltman, T.: Data value and the search for a single source of truth: What is it and why does it matter? In: Proceedings of the 57th Hawaii International Conference on System Sciences. HICSS, Hawaii International Conference on System Sciences (2024). <https://doi.org/10.24251/hicss.2024.795>, <http://dx.doi.org/10.24251/HICSS.2024.795>
36. Rakshit, S., et al.: geograpy3: Extract place names from text and resolve them to countries/regions. <https://github.com/somnathrakshit/geograpy3> (2024)
37. Ronzano, F., del Bosque, G.C., Saggion, H.: Semantify CEUR-WS Proceedings: Towards the Automatic Generation of Highly Descriptive Scholarly Publishing Linked Datasets, pp. 83–88. Springer International Publishing (2014). https://doi.org/10.1007/978-3-319-12024-9_10

38. Tkaczyk, D., Szostek, P., Fedoryszak, M., Dendek, P.J., et al.: CERMINE: automatic extraction of structured metadata from scientific literature. *International Journal on Document Analysis and Recognition (IJDAR)* **18**(4), 317–335 (Jul 2015). <https://doi.org/10.1007/s10032-015-0249-8>
39. Wiermann, B.: K10plus – Zehn Bundesländer in einem Bibliothekssystem (2019), <https://blog.slub-dresden.de/beitrag/2019/03/27/k10plus-zehn-bundeslaender-in-einem-bibliothekssystem>
40. WikiCite Contributors: WikiCite: A bibliographic database based on Wikidata. <http://wikicite.org/>
41. Wikidata Contributors: Requests for permissions/bot/ceur-ws (2023), https://www.wikidata.org/wiki/Wikidata:Requests_for_permissions/Bot/CEUR-WS
42. Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., et al.: The FAIR guiding principles for scientific data management and stewardship. *Scientific Data* **3**(1) (Mar 2016). <https://doi.org/10.1038/sdata.2016.18>
43. Willighagen, E.L., Mietchen, D., Patel-Schneider, P., Linden, K., et al.: Scholia 2026: compliance with SPARQL 1.1. <https://raw.githubusercontent.com/WolfgangFahl/ScholiaGraphSplitPaper/main/main.pdf> (2026), accepted for SWAT4HCLS 2026; CEUR-WS volume forthcoming (ISSN 1613-0073)
44. Yu, C., Zhang, C., Wang, J.: Extracting body text from academic PDF documents for text mining. *CoRR* **abs/2010.12647** (2020), <https://arxiv.org/abs/2010.12647>